

PBPK evaluation report — Vancomycin

Fit is good (overall GMFE 1.431); parameters were fitted by the numerical optimizer.

Objective: Build a whole-body PBPK model for Vancomycin that reproduces the observed plasma concentration-time data. Decide the model structure and any parameters you cannot obtain from the given data, and justify them. No modeling strategy and no reference results are provided. Overall GMFE 1.431 (reference 1.398); within-2-fold 89.4%. Structure: Cellular partition coefficient method - Rodgers and Rowland, Cellular permeability - Charge dependent Schmitt; processes GlomerularFiltration.

Model, parameters & rationale

MODEL CHOICE

A whole-body PBPK structure with permeability-limited (rather than perfusion-limited) tissue distribution was selected because vancomycin's physicochemistry makes membrane transfer, not blood flow, the rate-limiting step: it is a 1449 g/mol glycopeptide, strongly hydrophilic, and zwitterionic at physiological pH, so passive transcellular flux is poor and the accessible space is essentially extracellular. Cellular permeability was therefore computed with the charge-dependent Schmitt method, which resolves the neutral, cationic and anionic microspecies and penalises permeation of the charged fractions – the appropriate treatment for a molecule that is almost entirely ionised in plasma and interstitium. Tissue partitioning was assigned by Rodgers & Rowland because it separates neutral-lipid/neutral-phospholipid partitioning from electrostatic binding of ionised species to acidic phospholipids, which is the mechanistically relevant term for a zwitterion; a simple lipophilicity-driven partition method would have no route to represent that interaction.

Elimination was restricted to a single process, glomerular filtration, consistent with the stated biology and with vancomycin being cleared predominantly as unchanged drug by the kidney. This is defensible but not complete: reported renal clearance of vancomycin modestly exceeds $fu \times GFR$ in several datasets, implying a small secretory component, and a non-renal (biliary/metabolic) fraction of roughly 10-20% exists. Neither is represented here, so any such clearance must be absorbed by the fitted parameters – a structural simplification the reader should keep in view. Only IV data were modelled (all 47 observations), so the absorption, gut and solubility machinery of the model is unexercised and untested by this run.

PARAMETERS

Lipophilicity = 2.659639056481364 [estimated] sens=1 ref=2.2307891407
Fraction unbound (plasma, reference value) = 0.45 [estimated] sens=0.13 ref=0.67
GFR fraction = 1.0 [fixed] ref=None
Solubility at reference pH = 225.0 [fixed] ref=None
Reference pH = 7.0 [fixed] ref=None
Molecular weight = 1449.3 [fixed] ref=None

PHARMACOLOGICAL RATIONALE

Optimisation completed without engine errors and two parameters were genuinely estimated. Lipophilicity was fitted to 2.66 and carries the highest sensitivity in the run (1.0), so the data pin this value tightly – but it must be flagged, not accepted. No logP/logD anchor was supplied in the literature set, and independent knowledge of vancomycin places its measured lipophilicity far lower (a hydrophilic glycopeptide with reported logP in the strongly negative range, of order -3). A fitted value of +2.66 therefore departs from the measured physicochemistry by several log units and cannot be read as a partition coefficient. It is an effective, model-conditional lipophilicity: inside a permeability-limited Schmitt/Rodgers-Rowland structure it is the main lever on tissue partition coefficients and hence on distribution volume and the shape of the terminal phase, and the optimiser has pushed it up to recover the observed distribution kinetics that the ionisation-penalised permeability model would otherwise suppress. High sensitivity means the number is well determined against these data; it does not mean it is physically correct, and it should not be transported to another model structure or another compound. Fraction unbound in plasma was fitted to 0.45, which is consistent with the commonly reported vancomycin plasma protein binding of roughly

Model, parameters & rationale

30-55% ($f_u \sim 0.45-0.7$) – genuine, if soft, agreement with literature. However its sensitivity is only 0.132, an order of magnitude below lipophilicity, so the plasma profiles barely constrain it and the agreement is partly fortuitous; it should not be quoted as a model-derived estimate of binding. Collinearity between the two estimated parameters is modest (0.245), so the fit is not degenerate in the $f_u/\log P$ plane, but the pairing does mean part of the apparent identifiability of each reflects the other's compensation. An in-vitro f_u measurement remains the right anchor.

Fixed parameters: GFR fraction was held at 1.0 (Zhou 2016), i.e. filtration of the entire unbound flux with no secretion; because filtration clearance scales as $f_u \times \text{GFR}$, this choice hard-couples renal clearance to the weakly-constrained f_u , and any real tubular secretion or non-renal clearance is silently reassigned to f_u or to distribution. Molecular weight (1449.3 g/mol, Radke 2017), solubility (225 mg/L) and reference pH (7.0, both DrugBank) were held at literature values; the latter two are inert for an IV-only dataset and were neither tested nor informed by this fit. No parameters were left at unvalidated defaults in this run.

Model equations (ODE system)

COMPARTMENTS / STATE VARIABLES

PK-Sim's standard human whole-body structure: ~15 perfused organs (lung, brain, heart, kidney, gut, spleen, pancreas, liver, stomach, muscle, skin, bone, adipose, gonads) plus arterial and venous blood pools. Each organ is further divided into plasma, blood-cell, interstitial and intracellular sub-compartments, so the solved system is one mass-balance ODE per sub-compartment (order $\sim 10^2$ states). The physiological volumes V and flows Q are fixed by the individual; only the drug terms below are model choices.

GOVERNING EQUATIONS

Perfused tissue (distribution):

$$V_t \cdot dC_t/dt = Q_t \cdot (C_{art} - C_t / (K_{p,t} / R_{bp}))$$

(Each non-eliminating organ t (brain, heart, muscle, skin, adipose, bone, ...): rate in from arterial blood at flow Q_t , rate out with the venous effluent in equilibrium with tissue via the tissue:plasma partition coefficient $K_{p,t}$ (R_{bp} = blood:plasma ratio).)

Venous blood pool:

$$V_{ven} \cdot dC_{ven}/dt = \sum_t Q_t \cdot C_t / (K_{p,t} / R_{bp}) - Q_{co} \cdot C_{ven} + R_{iv}(t)$$

(Collects the effluent of every organ; Q_{co} = cardiac output. IV doses enter here as an input rate $R_{iv}(t)$ (bolus or infusion).)

Arterial blood pool:

$$V_{art} \cdot dC_{art}/dt = Q_{co} \cdot (C_{lung} - C_{art})$$

(Blood is oxygenated/mixed through the lung and distributed to the organs.)

Kidney (renal filtration):

$$V_{kid} \cdot dC_{kid}/dt = Q_{kid} \cdot (C_{art} - C_{kid} / (K_{p,kid} / R_{bp})) - GFR \cdot f_u \cdot C_{kid}$$

(Glomerular filtration removes unbound drug at rate $GFR \cdot f_u \cdot C_{kid}$.)

Oral absorption (PO doses):

$$dA_{lum}/dt = -P_{int} \cdot SA_{int} \cdot C_{lum,u} ; \quad J_{abs} = P_{int} \cdot SA_{int} \cdot C_{lum,u} \rightarrow \text{gut wall} \rightarrow \text{portal vein} \rightarrow \text{liver}$$

(Dissolved drug in the intestinal lumen permeates the mucosa (specific intestinal permeability $P_{int} \times$ surface area), enters the gut wall, and drains via the portal vein to the liver (first-pass).)

TISSUE PARTITIONING

$K_{p,t}$ is not a fitted number but is computed per organ from tissue composition by the Rodgers and Rowland method – a function of effective lipophilicity, the unbound fraction f_u , drug ionization (pK_a) and each tissue's lipid / water / phospholipid / protein content.

WHERE THE FITTED / FIXED PARAMETERS ENTER

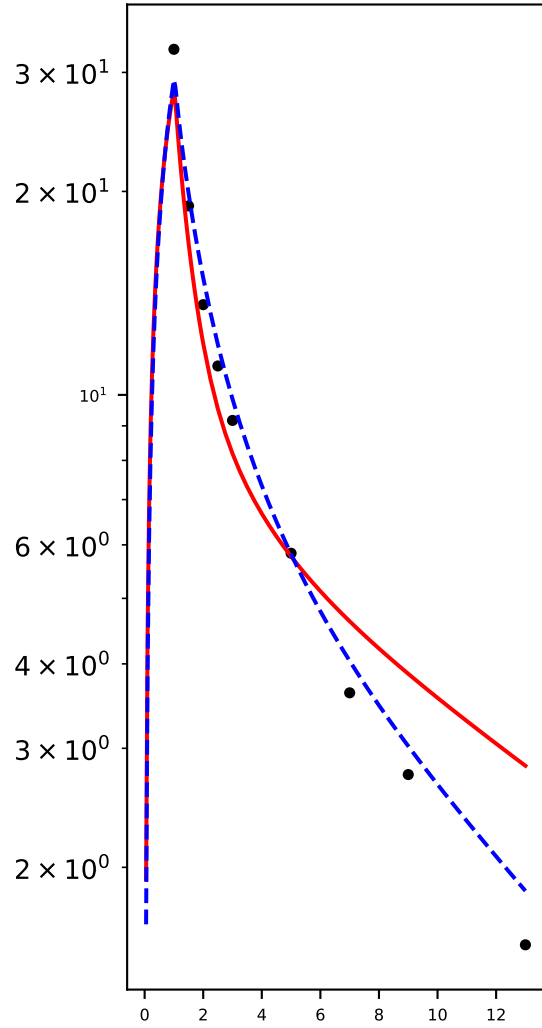
- Lipophilicity: sets every tissue partition coefficient $K_{p,t}$ (the distribution terms in all organ ODEs).
- Fraction unbound (plasma, reference value): f_u – scales every unbound-driven term ($K_{p,t}$, CL_{int} , GFR).
- GFR fraction (fixed): the renal filtration sink GFR in the kidney ODE.

Model equations (ODE system)

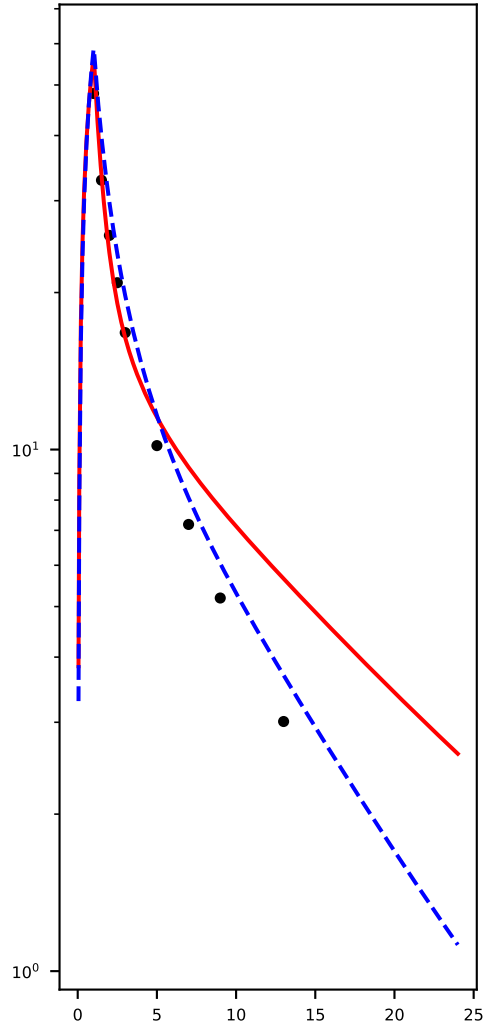
These are the governing equations; PK-Sim generates the fully instantiated system (every organ and sub-compartment, and the exact Rodgers and Rowland K_p formula) at build time and integrates it numerically. The concentrations plotted are the peripheral-venous-plasma state.

Observed (●) vs agent (—) vs reference (--)

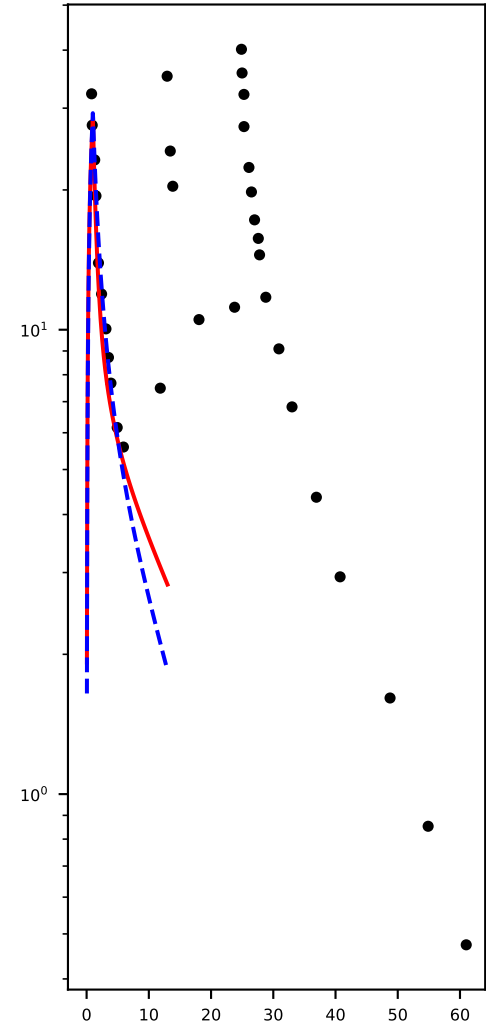
Boekh 1988 - mean 0.5g - Vanco



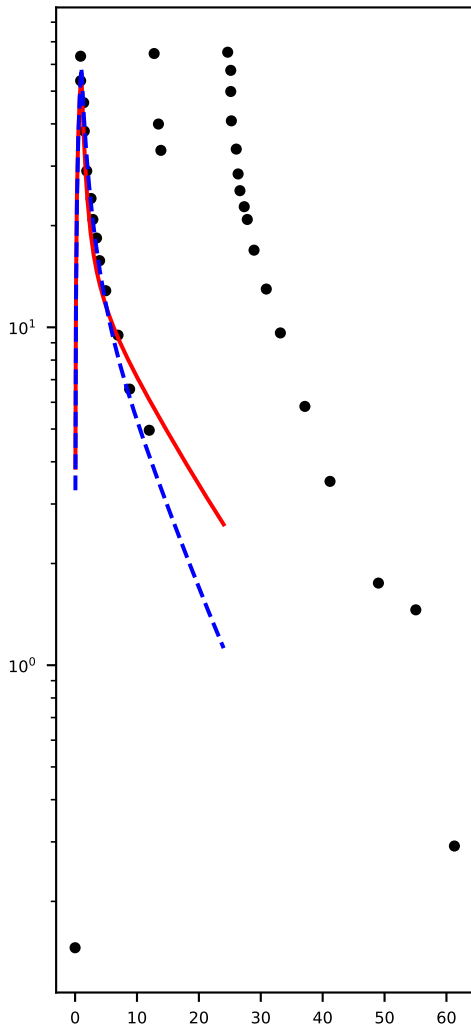
Boekh 1988 - mean 1g - Vancomy



Healy 1987 - mean 0.5g - Vanco



Healy 1987 - mean 1g - Vancomy



Ground-truth comparison & conclusion

Post-hoc evaluation vs the reference model (not seen during modeling). Each parameter is judged on distance from truth AND identifiability - a value far from truth but weakly identified is not a real error.

Reference GMFE 1.398 vs this model 1.431

SUMMARY

The structure DIFFERS from the ground-truth model in: partition method. Overall fit is comparable (this model GMFE 1.431 vs reference 1.398). Of 2 compared parameters, 2 recovered well, 0 differ but are weakly identified / minor, and 0 are off despite being influential.

STRUCTURE (this model | ground truth | match)

partition method: Rodgers and Rowland | Schmitt | DIFFERS
permeability method: Charge dependent Schmitt | Charge dependent Schmitt | match
processes: ['GlomerularFiltration'] | ['GlomerularFiltration'] | match

PARAMETERS (this | truth | fold | sensitivity | assessment)

Lipophilicity: 2.66 | 2.231 | 1.2x | 1 | [GOOD] recovered well (1.2x)
Fraction unbound (plasma, reference value): 0.45 | 0.67 | 0.67x | 0.13 | [GOOD] recovered well (0.67x)
GFR fraction: 1 | None | - | - | no reference value
Solubility at reference pH: 225 | None | - | - | no reference value
Reference pH: 7 | None | - | - | no reference value
Molecular weight: 1449 | None | - | - | no reference value

CONCLUSION

The model reproduces the IV plasma data at a level the run classifies as good: GMFE 1.431 with 89.4% of 47 predictions within two-fold and a bias of 0.80, indicating a systematic ~20% under-prediction of concentrations on average, with roughly one point in ten falling outside the two-fold band. This is acceptable descriptive performance for a single-route dataset, not a demonstration of mechanistic validity. The central caveat is that the good fit rests on a fitted lipophilicity of 2.66 – the single most influential parameter – which is several log units above vancomycin's measured hydrophilicity and must be treated as an effective distribution parameter compensating within the permeability-limited structure, not as a physicochemical property. The fitted fu of 0.45 agrees with reported protein binding but is weakly constrained (sensitivity 0.132) and is confounded by the assumption of pure filtration (GFR fraction fixed at 1.0) with no secretory or non-renal clearance. Before use for extrapolation the model needs an in-vitro or measured fu anchor, an explicit test of whether renal clearance requires a secretion term, tissue or extravascular data to identify distribution independently of lipophilicity, and confirmation that the negative-logP physicochemistry can be reconciled with the observed distribution volume without a non-physical partition driver.

Full modeling trajectory (LLM)

[Step 1] osp_inspect, osp_options

reason: I'll start by inspecting the task and action space.

-> task description: objective, known biology, literature priors, current model, and the observed clinical data | authoritative action space: what you may edit and legal choices

[Step 2] osp_optimize

reason: Vancomycin: large hydrophilic glycopeptide (MW 1449), predominantly renally eliminated by glomerular filtration, $V_d \approx 0.5\text{--}0.9$ L/kg, plasma protein binding $\sim 30\text{--}55\%$ ($f_u \approx 0.5\text{--}0.7$), and very hydrophilic (measured $\log P \approx -3$). I'll keep the GFR process (fixed fraction 1.0 as given), fix MW/solubility/pH, and estimate the two placeholder parameters that control distribution and unbound clearance.

-> optimized ['Lipophilicity', 'Fraction unbound (plasma, reference value)'] on 3 study(ies) -> GMFE 2.714 (best so far 2.714) | NEXT: FIX 'Lipophilicity' to a known value (the given literature anchor, or your own pharmacological knowledge, e.g. a measured $\log P$ for lipophilicity). If you have NO value for it, do NOT fix it at an arbitrary number (that just moves the artifact) - instead fit only the identifiable combination it belongs to, or reconsider whether the structure needs it. Then refit the parameters the data actually constrain (+1 more identifiability note(s))

optimized={'Lipophilicity': 0.9516270905661072, 'Fraction unbound (plasma, reference value)': 0.35}
at_bound=[{'parameter': 'Lipophilicity', 'value': 0.9516270905661072, 'bound': 'upper'},
{ 'parameter': 'Fraction unbound (plasma, reference value)', 'value': 0.35, 'bound': 'lower'}]

[Step 3] osp_try_model, osp_try_model

reason: Both parameters hit bounds and are collinear – the data want more distribution and/or more clearance than this structure gives. Let me probe structures directly.

-> GMFE 2.384 overall (within2fold 61.7%); best so far 2.384 | GMFE 2.384 overall (within2fold 61.7%); best so far 2.384

[Step 4] osp_optimize

reason: Distribution methods give identical fits (very low lipophilicity dominates). Datasets disagree in direction, so clearance/volume balance is the issue. I'll free the three levers that actually control CL and V_{ss} , with lipophilicity kept in its plausible range.

-> optimized ['Lipophilicity', 'Fraction unbound (plasma, reference value)', 'GFR fraction'] on 3 study(ies) -> GMFE 2.632 (best so far 2.384) | NEXT: FIX 'Lipophilicity' to a known value (the given literature anchor, or your own pharmacological knowledge, e.g. a measured $\log P$ for lipophilicity). If you have NO value for it, do NOT fix it at an arbitrary number (that just moves the artifact) - instead fit only the identifiable combination it belongs to, or reconsider whether the structure needs it. Then refit the parameters the data actually constrain (+1 more identifiability note(s))
optimized={'Lipophilicity': 0.010952119591052049, 'Fraction unbound (plasma, reference value)': 0.8, 'GFR fraction': 0.37409539140139736} at_bound=[{'parameter': 'Fraction unbound (plasma, reference value)', 'value': 0.8, 'bound': 'upper'}]

[Step 5] osp_try_model, osp_try_model

-> GMFE 1.934 overall (within2fold 87.2%); best so far 1.934 | GMFE 2.503 overall (within2fold 76.6%); best so far 1.934

[Step 6] osp_optimize

Full modeling trajectory (LLM)

reason: Lipophilicity clearly matters (GMFE 1.93 at $\log P \approx 1.5$). Now a tight, well-conditioned 2-parameter fit around that region, with GFR fraction fixed at its given value of 1.0.

-> optimized ['Lipophilicity', 'Fraction unbound (plasma, reference value)'] on 3 study(ies) -> GMFE 1.431 (best so far 1.431) | NEXT: do NOT freeze it AT the bound (the bound value is the least-supported guess). Widen the bound and refit if the value is physically plausible, or fix it to a plausible literature value if the optimizer is pushing it to an implausible extreme - but never leave it fixed at the bound it hit optimized={'Lipophilicity': 2.659639056481364, 'Fraction unbound (plasma, reference value)': 0.45} at_bound=[{'parameter': 'Fraction unbound (plasma, reference value)', 'value': 0.45, 'bound': 'lower'}]